

LEVEL 7 — Universal Functions (ufunc)

Easy line-by-line notes with examples

First line

```
import numpy as np
```

1. What is a ufunc?

A ufunc means Universal Function. It performs operations efficiently on NumPy array elements.

```
arr=np.array([1,4,9])
```

```
print(np.sqrt(arr))
```

Output: [1. 2. 3.]

2. Element-wise operations

The operation is performed separately on each element.

```
arr=np.array([1,4,9,16])
```

```
print(np.sqrt(arr))
```

Output: [1. 2. 3. 4.]

3. Unary ufunc

A unary ufunc takes one input.

```
arr=np.array([1,4,9])
```

```
result=np.sqrt(arr)
```

```
print(result)
```

4. Binary ufunc

A binary ufunc works with two inputs.

```
a=np.array([10,20,30])
```

```
b=np.array([1,2,3])
```

```
print(np.add(a,b))
```

Output: [11 22 33]

5. np.sqrt()

Square root of each element.

```
a=np.array([4,9,16,25])
```

```
print(np.sqrt(a))
```

Output: [2. 3. 4. 5.]

6. np.exp()

Calculates e raised to each value.

```
a=np.array([0,1,2])
```

```
print(np.exp(a))
```

Approximately: [1. 2.718 7.389]

7. np.log()

Natural logarithm, base e.

```
a=np.array([1,np.e,np.e**2])
```

```
print(np.log(a))
```

Approximately: [0. 1. 2.]

8. np.log10()

Base-10 logarithm.

```
a=np.array([1,10,100,1000])
```

```
print(np.log10(a))
```

Output: [0. 1. 2. 3.]

9. np.sin()

Sine of angles in radians.

```
a=np.array([0,np.pi/2,np.pi])
```

```
print(np.sin(a))
```

Approximately: [0. 1. 0.]

10. np.cos()

Cosine of angles in radians.

```
a=np.array([0,np.pi/2,np.pi])
```

```
print(np.cos(a))
```

Approximately: [1. 0. -1.]

11. np.tan()

Tangent of angles in radians.

```
a=np.array([0,np.pi/4])
```

```
print(np.tan(a))
```

Approximately: [0. 1.]

12. np.floor()

Rounds down to the nearest integer.

```
a=np.array([1.2,2.9,-1.2])
```

```
print(np.floor(a))
```

Output: [1. 2. -2.]

13. np.ceil()

Rounds up to the nearest integer.

```
a=np.array([1.2,2.1,-1.2])
```

```
print(np.ceil(a))
```

Output: [2. 3. -1.]

14. np.round()

Rounds to a chosen number of decimal places.

```
a=np.array([1.234,2.567,3.891])
```

```
print(np.round(a,2))
```

Output: [1.23 2.57 3.89]

15. np.trunc()

Removes the fractional part by moving toward zero.

```
a=np.array([1.9,2.7,-1.9,-2.7])
```

```
print(np.trunc(a))  
Output: [ 1.  2. -1. -2.]
```

16. np.absolute()

Returns absolute values; equivalent to np.abs().

```
a=np.array([-10, -5, 0, 5, 10])  
print(np.absolute(a))  
Output: [10  5  0  5 10]
```

Quick Revision

ufunc → Universal Function

Unary → one input

Binary → two inputs

Element-wise → operates on each element

sqrt → square root | exp → e^x | log → natural log | log10 → base-10 log

sin/cos/tan → trigonometric functions | floor → down | ceil → up

round → chosen decimals | trunc → toward zero | absolute → absolute value